```
.
├── node_modules
├── public/
│   ├── assets/
│   │   └── images.png
│   └── db.json
├── src/
│   ├── components/
│   │   ├── Footer.js
│   │   ├── Header.js
│   │   └── ProtectedRoute
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│   ├── pages/
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│   │   │   ├── Dashboard.js
│   │   │   ├── Profile.js
│   │   │   └── Settings.js
│   │   ├── Home.js
│   │   ├── Login.js
│   │   ├── ProductAdmin.js
│   │   ├── ProductDetail.js
│   │   └── Product.js
│   ├── router/
│   │   └── index.js
│   ├── App.js
│   └── index.js
├── .gitignore
├── package-lock.json
└── package.json
```

Index.js

```
import React from "react";
import ReactDOM from "react-dom/client";
import { BrowserRouter } from "react-router-dom";
import App from "./App";// src/index.js
import 'bootstrap/dist/css/bootstrap.min.css';
import 'bootstrap/dist/js/bootstrap.bundle.min.js'; // si usarás componentes con JS
(collapse, modal, etc.)
import './index.css';

const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(
 <BrowserRouter>
  <App />
 </BrowserRouter>
);
```

App.js

```
import Header from "./components/Header";
import AppRouter from "./router";
import Footer from "./components/Footer"; // si lo tienes de prácticas previas
```

```
export default function App() {
 return (
  <>
   <Header />
   <AppRouter />
   {typeof Footer === "function" && <Footer />}
  </>
 );
}
```

AppRouter.js

```
import { Routes, Route, Navigate } from "react-router-dom";
// Páginas
import Home from "../pages/Home";
import Products from "../pages/Products";
import ProductDetail from "../pages/ProductDetail";
import ProductAdmin from "../pages/ProductAdmin"
import Login from "../pages/Login";

// Dashboard (rutas anidadas)
import Dashboard from "../pages/Dashboard/Dashboard";
import Profile from "../pages/Dashboard/Profile";
import Settings from "../pages/Dashboard/Settings";

// Rutas protegidas
import ProtectedRoute from "../components/ProtectedRoute";

export default function AppRouter() {
 return (
  <Routes>
   <Route path="/" element={<Home />} />
   <Route path="/products" element={<Products />} />
   <Route path="/products/:id" element={<ProductDetail />} />
   <Route path="/admin/products" element={<ProductAdmin />} />
   <Route path="/login" element={<Login />} />

   {/* Rutas anidadas protegidas */}
   <Route
    path="/dashboard"
    element={(
     <ProtectedRoute>
      <Dashboard />
     </ProtectedRoute>
    )}
   >
    <Route path="profile" element={<Profile />} />
    <Route path="settings" element={<Settings />} />
   </Route>

   {/* Redirección de rutas antiguas */}
   <Route path="/old-products" element={<Navigate to="/products" replace />} />

   {/* 404 */}
   <Route path="*" element={<h2>Página no encontrada</h2>} />
  </Routes>
 );}
```

useLocalDB.js

```
import { useEffect, useMemo, useState } from "react";

const STORAGE_KEY = "productos_db_v1";

export function useLocalDB() {
 const [data, setData] = useState([]);     // Array de productos
 const [loading, setLoading] = useState(true);

 // Carga inicial desde localStorage o desde /products.json
 useEffect(() => {
  try {
   const stored = localStorage.getItem(STORAGE_KEY);
   if (stored) {
    setData(JSON.parse(stored));
    setLoading(false);
    return;
   }
  } catch (_) {
   // Si parse falla, seguimos y recargamos desde JSON
  }
  fetch(process.env.PUBLIC_URL + "/products.json")
   .then(function (res) { return res.json(); })
   .then(function (json) {
    const productos = Array.isArray(json) ? json : (json && json.productos) ?
json.productos : [];
    localStorage.setItem(STORAGE_KEY, JSON.stringify(productos));
    setData(productos);
   })
   .catch(function () {
    // Si falla la carga inicial, dejamos data como []
    setData([]);
   })
   .finally(function () {
    setLoading(false);
   });
 }, []);

 // Guardar cambios en memoria + persistencia
 function save(newData) {
  setData(newData);
  localStorage.setItem(STORAGE_KEY, JSON.stringify(newData));
 }
 // Generador de IDs sencillo, sin backticks ni crypto
 function genId() {
  return "prod-" + Math.random().toString(36).slice(2, 10);
 }
 // CRUD
 function create(item) {
  const trimmed = (item && item.id) ? String(item.id).trim() : "";
  const newId = trimmed !== "" ? trimmed : genId();
  const newItem = { ...item, id: newId };
  save([ ...data, newItem ]);
  return newItem;
 }
```

```
 function update(id, updates) {
  const next = data.map(function (p) {
   return (p.id === id) ? { ...p, ...updates } : p;
  });
  save(next);
 }

 function remove(id) {
  const next = data.filter(function (p) { return p.id !== id; });
  save(next);
 }

 // Lectura
 function findById(id) {
  return data.find(function (p) { return p.id === id; });
 }

 function search(q) {
  const t = (q || "").toLowerCase();
  return data.filter(function (p) {
   return (
    String(p.id).toLowerCase().includes(t) ||
    String(p.nombre).toLowerCase().includes(t) ||
    String(p.precio).includes(t)
   );
  });
 }
 return useMemo(function () {
  return { data, loading, create, update, remove, findById, search };
 }, [data, loading]);
}
```

Header.js

```
import { NavLink } from "react-router-dom";

export default function Header() {
 const link = ({ isActive }) =>
  "nav-link" + (isActive ? " fw-bold text-decoration-underline" : "");

 return (
  <header className="bg-dark text-white p-3 mb-4">
   <div className="container d-flex align-items-center justify-content-between">
    <h1 className="h4 mb-0">Tienda de Productos</h1>
    <nav>
     <ul className="nav">
      <li className="nav-item"><NavLink to="/" className={link}>Inicio</NavLink></li>
      <li className="nav-item"><NavLink to="/products"
className={link}>Productos</NavLink></li>
     <li className="nav-item"><NavLink to="/admin/products" className={link}>Admin
Productos</NavLink></li>
      <li className="nav-item"><NavLink to="/dashboard"
className={link}>Dashboard</NavLink></li>
      <li className="nav-item"><NavLink to="/login" className={link}>Login</NavLink></li>
     </ul>
    </nav>
```

```
   </div>
  </header>
 );
}
```

ProtectedRoute.js

```
import { Navigate } from "react-router-dom";

export default function ProtectedRoute({ children }) {
 const isAuth = localStorage.getItem("token");
 return isAuth ? children : <Navigate to="/login" replace />;
}
```

Login.js

```
import { useNavigate } from "react-router-dom";

export default function Login() {
 const navigate = useNavigate();

 const handleLogin = () => {
  localStorage.setItem("token", "token-demo");
  navigate("/dashboard", { replace: true });
 };

 const handleLogout = () => {
  localStorage.removeItem("token");
 };

 return (
  <section className="container py-4">
   <h1>Login</h1>
   <button className="btn btn-primary me-2" onClick={handleLogin}>
    Iniciar sesión (demo)
   </button>
   <button className="btn btn-outline-secondary" onClick={handleLogout}>
    Cerrar sesión
   </button>
  </section>
 );
}
```

Dashboard.js

```
import { Link, Outlet } from "react-router-dom";

export default function Dashboard() {
 return (
  <section className="container py-4">
   <h1>Panel de Control</h1>
   <nav className="d-flex gap-3 mb-3">
    <Link to="profile">Perfil</Link>
    <Link to="settings">Configuración</Link>
   </nav>
   <Outlet />
```

```
  </section>
 );
}
```

Products.js

```
import { Link } from "react-router-dom";
import { useState } from "react";
import { useLocalDB } from "../hooks/useLocalDB";

export default function Products() {
 const { data, loading, search, remove } = useLocalDB();
 const [query, setQuery] = useState("");
 const items = query ? search(query) : data;
 if (loading) return <p className="container py-4">Cargando…</p>;

 return (
  <section className="container py-4">
   <h1 className="mb-3">Productos</h1>

   <div className="input-group mb-3">
    <input
     className="form-control"
     placeholder="Buscar por id, nombre o precio…"
     value={query}
     onChange={(e) => setQuery(e.target.value)}
    />
    {query && (
     <button className="btn btn-outline-secondary" onClick={() =>
setQuery("")}>Limpiar</button>
    )}
   </div>

   <div className="row g-3">
    {items.map((p) => (
     <div key={p.id} className="col-12 col-md-6 col-lg-4">
      <div className="card h-100">
       {p.imagen && (
        <img src={p.imagen} className="card-img-top" alt={p.nombre} />
       )}
       <div className="card-body">
        <h5 className="card-title">{p.nombre}</h5>
        <p className="card-text mb-1"><strong>Precio:</strong> {p.precio.toFixed(2)} € </p>
        <p className="card-text">
         <span className={`badge ${p.enStock ? "bg-success" : "bg-danger"}`}>
          {p.enStock ? "En stock" : "Sin stock"}
         </span>
        </p>
        <div className="d-flex gap-2">
         <Link className="btn btn-primary btn-sm" to={`/products/${p.id}`}>Ver
detalle</Link>
         <button
          className="btn btn-outline-danger btn-sm"
          onClick={() => remove(p.id)}>Eliminar</button>
        </div>
       </div>
      </div>
```

```
    </div>))}
    {items.length === 0 && <p>No hay resultados.</p>}
   </div>
  </section>
 );
}
```

ProductsAdmin.js

```
import { useState } from "react";
import { useLocalDB } from "../hooks/useLocalDB";

const empty = { id: "", nombre: "", precio: 0, enStock: true, imagen: "", descripcion:
"" };

export default function ProductAdmin() {
 const { data, create, update, remove } = useLocalDB();
 const [form, setForm] = useState(empty);
 const [editId, setEditId] = useState(null);

 const handleSubmit = (e) => {
  e.preventDefault();
  const precio = Number(form.precio);
  if (!form.nombre.trim() || Number.isNaN(precio) || precio <= 0) return;

  if (editId) {
   update(editId, { ...form, precio });
  } else {
   create({ ...form, precio });
  }
  setForm(empty);
  setEditId(null);
 };

 const startEdit = (p) => {
  setEditId(p.id);
  setForm(p);
 };

 return (
  <section className="container py-4">
   <h1>Administrar productos</h1>

   <form className="row g-3 mb-4" onSubmit={handleSubmit}>
    <div className="col-md-3">
     <label className="form-label">ID (opcional si creas)</label>
     <input className="form-control" value={form.id} onChange={(e) => setForm({ ...form,
id: e.target.value })} />
    </div>
    <div className="col-md-3">
     <label className="form-label">Nombre</label>
     <input className="form-control" value={form.nombre} onChange={(e) => setForm({
...form, nombre: e.target.value })} required />
    </div>
    <div className="col-md-2">
     <label className="form-label">Precio (€)</label>
     <input type="number" step="0.01" min="0" className="form-control" value={form.precio}
```

```
onChange={(e) => setForm({ ...form, precio: e.target.value })} required />
    </div>
    <div className="col-md-2 d-flex align-items-end">
     <div className="form-check">
      <input id="chkStock" type="checkbox" className="form-check-input"
checked={form.enStock} onChange={(e) => setForm({ ...form, enStock:
e.target.checked })} />
      <label htmlFor="chkStock" className="form-check-label">En stock</label>
     </div>
    </div>
    <div className="col-12">
     <label className="form-label">Imagen (URL pública) y descripción</label>
     <input className="form-control mb-2" placeholder="/assets/tu-imagen.jpg"
value={form.imagen} onChange={(e) => setForm({ ...form, imagen: e.target.value })} />
     <textarea className="form-control" rows={2} value={form.descripcion} onChange={(e) =>
setForm({ ...form, descripcion: e.target.value })} />
    </div>
    <div className="col-12 d-flex gap-2">
     <button className="btn btn-success" type="submit">{editId ? "Guardar cambios" :
"Añadir producto"}</button>
     <button className="btn btn-secondary" type="button" onClick={() => { setForm(empty);
setEditId(null); }}>Reset</button>
    </div>
   </form>

   <div className="table-responsive">
    <table className="table table-striped align-middle">
     <thead>
      <tr>
       <th>ID</th><th>Nombre</th><th>Precio</th><th>Stock</th><th style={{width: 220}}
>Acciones</th>
      </tr>
     </thead>
     <tbody>
      {data.map((p) => (
       <tr key={p.id}>
        <td>{p.id}</td>
        <td>{p.nombre}</td>
        <td>{p.precio.toFixed(2)} €</td>
        <td><span className={`badge ${p.enStock ? "bg-success" : "bg-danger"}`}
>{p.enStock ? "Sí" : "No"}</span></td>
        <td className="d-flex gap-2">
         <button className="btn btn-sm btn-outline-primary" onClick={() => startEdit(p)}
>Editar</button>
         <button className="btn btn-sm btn-outline-danger" onClick={() => remove(p.id)}
>Eliminar</button>
        </td>
       </tr>
      ))}
      {data.length === 0 && <tr><td colSpan={5}>Sin productos</td></tr>}
     </tbody>
    </table>
   </div>
  </section>
 );
}
```

ProductDetail.js

```
import { useParams, Link } from "react-router-dom";
import { useLocalDB } from "../hooks/useLocalDB";

export default function ProductDetail() {
 const { id } = useParams();
 const { findById } = useLocalDB();
 const product = findById(id);

 if (!product) {
  return (
   <section className="container py-4">
    <p>Producto no encontrado.</p>
    <Link to="/products">Volver al listado</Link>
   </section>
  );
 }

 return (
  <section className="container py-4">
   <h1>{product.nombre}</h1>
   {product.imagen && (
    <img
     src={product.imagen}
     alt={product.nombre}
     className="img-fluid mb-3"
     style={{ maxWidth: 320 }}
    />
   )}
   <p><strong>Código:</strong> {product.id}</p>
   <p><strong>Precio:</strong> {product.precio.toFixed(2)} €</p>
   <p><strong>Disponible:</strong> {product.enStock ? "Sí" : "No"}</p>
   {product.descripcion && <p>{product.descripcion}</p>}
   <Link to="/products" className="btn btn-secondary mt-3">Volver</Link>
  </section>
 );
}
```

products.json

```
{
 "productos": [
  {
   "id": "lap-001",
   "nombre": "Lenovo ThinkPad E14",
   "precio": 799.99,
   "enStock": true,
   "imagen": "/assets/lenovothinkPadE14.jpg",
   "descripcion": "Portátil profesional 14"
  },
  {
   "id": "mon-101",
   "nombre": "LG UltraGear 27 144Hz",
   "precio": 279.0,
   "enStock": true,
```

```
   "imagen": "/assets/LGUltraGear27.jpg",
   "descripcion": "Monitor gaming 27 144Hz"
  },
  {
   "id": "per-501",
   "nombre": "Logitech MX Keys",
   "precio": 109.0,
   "enStock": false,
   "imagen": "/assets/LogitechMXKeys.jpg",
   "descripcion": "Teclado inalámbrico retroiluminado"}]}
```

SearchBar.js

```
import React from 'react';
import { useRef } from 'react';
export default function SearchBar({
  query,
  onChange,
  onClear,
  placeholder = 'Buscar libros por código, nombre, autor o precio',
}) {
  const handleInput = (e) => onChange(e.target.value);
  const handleClear = () => onClear?.();
  const inputRef = useRef();
  const enfocar = () => {inputRef.current.focus();};
  return (
    <div className="container mt-3">
      <div className="row justify-content-center">
        <div className="col-12 col-md-10 col-lg-8">
          <div className="input-group">
            <button className="btn btn-info text-light fw-bold border border-1 border-info"
id="search-label" onClick={enfocar}>  Buscar</button>
            <input
              ref={inputRef}
              type="text"
              value={query}
              onChange={handleInput}
              className="form-control border border-info"
              placeholder={placeholder}
              aria-label="Barra de búsqueda de libros"
              aria-describedby="search-label"
            />
            {query && (
              <button
                type="button"
                className="btn btn-outline-secondary"
                onClick={handleClear}
                title="Limpiar búsqueda">Limpiar</button>
            )}
          </div>
          <small className="text-dark d-block mt-1">
            Filtra en tiempo real por Codigo / Nombre / Autor / Precio. Ejemplos:
<code>“nb-001”</code>, <code>“Crimen y castigo”</code>, <code>“Harlan Ellison”</code>,
<code>“20.85”</code>.
          </small>
        </div>
      </div></div>);}
```

ProductList +SearchBar.js

```
import React, { useState } from "react";
// Componentes
import Header from "./components/Header";
import Footer from "./components/Footer";
import ProductList from "./components/ProductList";
import SearchBar from "./components/SearchBar";
import Cart from "./components/Cart";
// Datos
import { catalog } from "./data/catalog";
export default function App() {
 // Estado de búsqueda
 const [query, setQuery] = useState("");
 // Estado del carrito
 const [cartItems, setCartItems] = useState([]);
 // Añadir producto al carrito
 const addToCart = (product) => {
  setCartItems([...cartItems, product]);
 };
 // Filtrado en tiempo real
 const filteredProducts = catalog.filter((p) => {
  const text = query.toLowerCase();
  return (
   p.nombre.toLowerCase().includes(text) ||
   p.id.toLowerCase().includes(text) ||
   p.autor.toLowerCase().includes(text) ||
   String(p.precio).includes(text)
  );
 });
 return (
  <>
   {/* Cabecera */}
   <Header />
   {/* Carrito */}
   <div className="container mt-3">
    <Cart items={cartItems} />
   </div>
   {/* Barra de búsqueda */}
   <SearchBar
    query={query}
    onChange={setQuery}
    onClear={() => setQuery("")}
   />
   {/* Listado filtrado + botón Añadir */}
   <ProductList
    products={filteredProducts}
    onAdd={addToCart}
   />
   {/* Pie de página */}
   <Footer />
  </>
 );
}
```

Cart.js

Parent.js

```
...
 const [cartItems, setCartItems] = useState([]);
 const addToCart = (product) => {
  setCartItems([...cartItems, product]);
 };
...
<Cart items={cartItems} />
...
export default function Cart({ items = [] }) {
  const total = items.length;
  let suma = 0;
  items.forEach((libro) => {
        suma += libro.precio;
        });
  return (
    <div className="cart container d-flex justify-content-around align-items-center">
      <span className="border border-2 border-info  text-dark p-2 rounded-3 ">
        Carrito ({total} {total === 1 ? 'libro' : 'libros'})
      </span>
      <div className='bg-info border border-2 border-info text-light p-2 rounded-3 text-
center'>
        <span className='fw-bold'>Precio final: {Math.round(suma * 100) / 100} €</span>
      </div>

    </div>
  )
};
```

Regex(s)

email	^[\w-\.]+@([\w-]+\.)+[\w-]{2,4}$
tel	^[\+]?[(]?[0-9]{3}[)]?[-\s\.]?[0-9]{3}[-\s\.]?[0-9]{4,6}$
First Last name	\b([A-ZÀ-ÿ][-,a-z. ']+[ ]*)+